

RGR (P47804) — Function-Assignment

Review: GPCR coupling, phototransduction, and plasma-membrane activity

Focus: function_assignment · **Hypothesis slug:** g-protein-coupling-phototransduction-and-active-membrane-scope **Gene:** human RGR (RPE-retinal G protein-coupled receptor), UniProt P47804 · **Iteration:** 1 of 3

Executive Judgment

Verdict: Over-annotated (refuted for the G-protein/signaling claims; partially/weakly supported only in a narrow, isoform-specific sense for plasma membrane).

The seed hypothesis proposes that human RGR *enables GPCR/photoreceptor activity, participates in GPCR signaling and phototransduction, and is active in the plasma membrane*. Evaluating each claim separately against primary literature, exact GO scope, and a direct sequence check:

1. **GPCR activity (GO:0004930) / G-protein-coupled photoreceptor activity (GO:0008020) — NOT SUPPORTED.** These MF terms require GDP/GTP exchange on a Gα subunit. Thirty years of research have produced **no coupled G protein and no positive coupling assay** for RGR

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RGR's demonstrated molecular function is **light-driven retinaldehyde photoisomerase** activity (atRAL→11cRAL)

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A direct scan of P47804 shows it **lacks both canonical class-A activation motifs** (no DRY/ERY — it has GRY at 112; no NPxxY — it has NAINY at 261), i.e., the structural determinants of canonical G-protein engagement are absent. A comparative scan (Iteration 3; see table below) shows RGR is the **only** human opsin sampled to have lost both — rhodopsin, red/blue cone opsins, and even peropsin all retain DRY and NPxxY — establishing this as target-specific divergence rather than a shared opsin-family trait. These are IBA terms propagated from the opsin ancestral node (PTN008398221) and represent **paralog/ancestral over-propagation**, not a target-specific property.

2. **GPCR signaling pathway (GO:0007186) — NOT SUPPORTED** for the same reason; no downstream heterotrimeric G-protein cascade has been demonstrated for RGR.
3. **Phototransduction (GO:0007602) — QUALIFIED / non-core.** GO:0007602 ("conversion of absorbed photons into a molecular signal") does not itself demand a heterotrimeric G protein. RGR does perform a **light-driven molecular event** (photoisomerization producing 11-cis-retinal, the chromophore that *enables* photoreceptor phototransduction). However, RGR acts in the **visual/retinoid cycle in RPE and Müller glia**, upstream of and physically separate from the photoreceptor phototransduction cascade — it supplies chromophore rather than transducing the light signal that leads to neuronal excitation. Assigning canonical "phototransduction" to RGR conflates chromophore regeneration with signal transduction; a **visual-cycle / chromophore-biosynthesis framing is more accurate** than GO:0007602 as a core BP.
4. **Plasma membrane (GO:0005886) — WEAKLY / ISOFORM-SPECIFICALLY SUPPORTED, not for the canonical product.** Full-length (normal) RGR is **ER/microsomal-membrane associated**

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RPE microsomes). Plasma-membrane localization is documented **only for the human exon-VI-skipping splice isoform RGR-d**

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which escapes ER-associated degradation, traffics to the RPE basolateral membrane, and **co-localizes with terminal complement complex C5b-9/vitronectin in Bruch's-membrane deposits** (an aging/AMD, likely pathological context). No functional signaling activity has

been shown at that membrane. This supports a *gene-level* CC of plasma membrane at most as isoform-specific, and does **not** justify an `is_active_in plasma membrane` (active MF at that location).

Most important caveats: (a) Absence of a positive coupling assay plus loss of DRY/NPxxY is strong convergent inference but is not a formal, controlled negative G-protein assay; (b) the non-ocular lead (RGR in human skin keratinocytes,

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PMC9014095) was read in full (Iteration 2) and does **not** rescue the signaling claim: the authors themselves call RGR "a photoisomerase," RGR localizes predominantly to **nucleus, mitochondria and cytoplasm** (immunoEM) rather than a canonical plasma-membrane GPCR pattern, and the only functional data are an **siRNA-knockdown phenotype** (proliferation/migration/apoptosis) with no G-protein coupling assay. This is a loss-of-function cellular phenotype, not evidence of G-protein activation.

Evidence Matrix

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confidence/ Limitations
39961453 (2025)	Review (synthesis, 30 yr)	Refutes G-protein claim	Does RGR couple a G protein?	"While named a G protein-coupled receptor, RGR has no known coupled G protein"; acts as alternative visual-cycle photoisomerase	Human/ vertebrate RPE & Müller glia	Highly negative state review
32493732 (2021)	Review of primary genetic/ biochemical work	Refutes GPCR-MF; supports photoisomerase	Core molecular function	RGR is a mammalian retinaldehyde photoisomerase homologous to invertebrate retinochrome; physiological relevance in 11cRAL regeneration established	Mammalian eye	Highly positive is to retinal photoreceptors not to opsins
37585292 (2023)	Primary (mouse ERG, cell-specific reactivation)	Supports photoisomerase BP	Physiological role	RGR is a pan-retinal atRAL sink supporting rapid chromophore regeneration in RPE + specialized Müller glia	Mouse retina	Highly positive photoreceptors not a assay
37057907 (2023)	Primary biophysics	Supports photoisomerase MF	Reaction mechanism	Reversible photoreaction of RGR as a retinal photoisomerase	In vitro	Highly positive isomerase mechanism

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confidence/ Limitations
18474598 (2008)	Primary (KO mice, RNAi human RPE, in vitro)	Qualifies	Signaling mechanism of light response	RGR mediates light-dependent retinyl-ester mobilization; in the dark RGR-opsin inhibits LRAT and atRE hydrolase in vitro, released by light. No G protein identified; mechanism left open	Human RPE cells + mouse	Highly phenotypic explanation not a G-protein mechanism
8639565 (1996)	Primary biochemistry (bovine + human seq)	Supports retinal-binding MF; localizes to microsomes	Chromophore binding; location	RGR binds all-trans-retinal forming pH-dependent photopigments; isolated from bovine RPE microsomes	Bovine RPE (+ human sequence)	Highly binding human = microsome ER.
7947717 (1994)	Primary biochemistry + structural/ evolutionary	Supports isomerase MF; qualifies GPCR framing	Retinal binding; family placement	Bovine RGR covalently binds all-trans- (preferentially) and 11-cis-retinal; human 86% identical by sequence; RGR is "the earliest independent branch of the vertebrate opsin gene family";	Bovine RPE protein + human gene/ sequence	Highly binding bovine by human evolution divergent argument opsin G-protein inheritance

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Conf. limit
				proposes retinochrome-like isomerization		
27011730 (2016)	Primary localization (human tissue, fetal RPE, splice-specific Ab)	Qualifies plasma-membrane claim	Subcellular location	RGR-d (exon-VI-skip isoform), not normal RGR, traffics to RPE basolateral plasma membrane; co-localizes with C5b-9/vitronectin in basal deposits	Human RPE, young vs old donors	Highly specific disease-associated activation of PM.
P47804 sequence scan (this analysis)	Computational/ structural	Refutes canonical coupling	Presence of GPCR activation motifs	No DRY/ERY (GRY@112); no NPxxY (NAINY@261) — both G-protein-coupling determinants absent	UniProt human RGR, 291 aa	Model inferred loss = negative
35445026 (2022)	Primary (skin/ keratinocytes; localization + knockdown)	Qualifies (does not support signaling)	Non-ocular signaling role	Authors call RGR "a photoisomerase"; RGR localizes to nucleus/ mitochondria/ cytoplasm (immunoEM) + membrane; siRNA knockdown alters	Human keratinocytes, melanocytes, fibroblasts	Model of function phenotypes intra-nuclear localization argued classically signaling

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Conf limit
				proliferation/ migration/ apoptosis. No G-protein coupling assay ; localization not canonical PM-GPCR		
8639565 + phylogeny (41618564, 2026)	Structural/ evolutionary	Supports isomerase framing	Family placement	RGRs are blue-absorbing rhodopsins acting as retinal photoisomerases across vertebrates	Cross-species	Supp cons phot not s

Comparative Motif Analysis (computed provenance, Iteration 3)

Direct scan of UniProt sequences for the two class-A GPCR activation motifs. **RGR is the only opsin sampled that has lost BOTH** — even peropsin (another non-visual opsin homolog) retains them:

UniProt	Protein	Length	DRY/ERY ([DE]R[YFW], TM3 ionic lock)	NPxxY (TM7)	G-protein coupling
P47804	RGR (human)	291	ABSENT (GRY@112)	NAINY@261 (Pro→Ala)	none demonstrated
P08100	Rhodopsin	348	ERY@134	NPVIY@302	Gt (transducin)
P04000	Red cone opsin	364	ERW@150	NPVIY@318	Gt
P03999	Blue cone opsin	345	ERY@128	NPIIY@296	Gt
O14718	Peropsin (RRH)	337	DRY@122	NPCIY@290	opsin homolog

Interpretation: loss of both motifs is specific to RGR, not a shared opsin-family trait, supporting **target-specific divergence** rather than inheritance of ancestral opsin G-protein function. (Computational inference; converges with the absence of any positive coupling assay in 30 years of literature.)

GO Curation Implications (leads — require curator verification)

GO term	Aspect	Current basis	Recommended action	Rationale
GO:0004930 G-protein coupled receptor activity	MF	IBA from opsin ancestor PTN008398221	Remove / do not propagate (NOT qualifier candidate)	No coupled G protein, no positive assay, DRY+NPxxY both absent. Ancestral over-propagation.
GO:0008020 G-protein coupled photoreceptor activity	MF	IBA	Remove / do not propagate	Requires Gα GDP/GTP exchange; unsupported. Target has diverged from opsin ancestor.
GO:0007186 GPCR signaling pathway	BP	IBA	Remove / do not propagate	No demonstrated heterotrimeric cascade.
GO:0007602 phototransduction	BP	IBA	Generalize / replace with visual-cycle framing (e.g., visual perception / retinoid metabolic process / chromophore regeneration). Treat as non-core for phototransduction sensu stricto	RGR regenerates chromophore in RPE/Müller glia; upstream of and separate from photoreceptor signal transduction.
GO:0005886 plasma membrane	CC	IBA / isoform	Retain only as isoform-specific (RGR-d), not <code>is_active_in</code>; prefer ER membrane (GO:0005789) for canonical product	Full-length RGR is ER/microsomal; PM localization is the RGR-d splice variant in a pathological deposit context.
Positive MF to consider adding	MF	Primary literature	Lead: retinal photoisomerase activity / all-trans-	Directly supported

GO term	Aspect	Current basis	Recommended action	Rationale
				(P 32493732 P 37057907 P 8639565). More informative than "protein binding".
Positive BP to consider	BP	Primary literature	Lead: visual cycle / retinoid metabolic process; regeneration of 11-cis-retinal	P 37585292 P 18474598 P 33187985 .

The lead recommendation is to **treat the five opsin-ancestor-propagated GPCR/phototransduction/plasma-membrane terms as over-annotations for canonical RGR** and to anchor the review on RGR's directly evidenced **photoisomerase/retinaldehyde-binding** function in the **RPE/Müller-glia visual cycle**, with plasma-membrane restricted to the RGR-d isoform.

Mechanistic Scope

- **Direct gene-product activity:** light absorption by RGR-bound all-trans-retinal (blue/near-UV photopigment) → **photoisomerization to 11-cis-retinal** (photoisomerase; MF). Also **retinaldehyde (all-trans-retinal) binding via a Schiff base**.
- **Immediate cellular role:** supplies 11-cis-retinal chromophore for opsin regeneration under photopic conditions in RPE and specialized Müller glia (auxiliary/photopic visual cycle).
- **Downstream / not direct RGR MF:** regeneration of photoreceptor light sensitivity, phototransduction in rods/cones, and the whole-visual-cycle flux are *consequences* of chromophore supply, not RGR's own signal-transduction activity.
- **Regulatory phenotype**

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light-dependent release of RGR's dark inhibition of LRAT and atRE hydrolase — a genuine light-regulated regulatory effect, but **no G-protein mediator identified**; mechanism explicitly left open.

- **Disease/pleiotropic:** RGR mutations → retinitis pigmentosa / peripapillary choroidal atrophy / AMD associations; RGR-d in Bruch's-membrane complement deposits — these are phenotypes, not evidence of a signaling MF.

Conflicts and Alternatives

- **Paralog/ancestral over-propagation:** the GPCR/phototransduction terms trace to opsin ancestral node PTN008398221 (IBD). RGR is a *retinochrome-type* photoisomerase, a functional divergence from signaling opsins; the target descends from the ancestor but has **diverged in function** (motif loss). Inheritance should not be assumed.
- **Historical naming artifact:** "RPE-retinal G protein-coupled receptor" is a homology-based name from the 1990s, repeatedly flagged in reviews as **not reflecting a demonstrated G-protein function**
([P 39961453](#)).
- **Isoform-specific localization:** plasma-membrane data are for RGR-d (exon-VI skip), which the canonical UniProt entry/annotation may not represent; using it to support gene-level PM `is_active_in` conflates isoforms.
- **Organism caveat:** foundational biochemistry (retinal binding, spectra) is **bovine** RGR
([P 8639565](#))
with human sequence context — not a direct human coupling experiment.
- **Non-ocular lead resolved (Iteration 2):** the keratinocyte study
([P 35445026](#))
was read in full. It does not support a G-protein/GPCR signaling capacity — RGR is again described as a photoisomerase, localizes mainly to nucleus/mitochondria/cytoplasm, and the functional readout is an siRNA-knockdown phenotype (proliferation/migration/apoptosis) with no coupling assay. If anything it suggests a non-canonical intracellular role, not plasma-membrane GPCR signaling.

• Evolutionary divergence

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RGR is the earliest-diverging branch of the vertebrate opsin family with a proposed retinochrome-like mechanism — direct support that opsin-ancestor (PTN008398221) G-protein signaling functions should not be assumed to propagate to RGR (target divergence, not mere donor-count).

Knowledge Gaps

1. Is there any positive or controlled-negative G-protein coupling assay for RGR?

Checked: no positive assay found; review explicitly says no coupled G protein. Matters because MF terms hinge on it. Resolve with a GTPγS/BRET/TGFα-shedding coupling panel (all Gα families) using purified/reconstituted human RGR ± light.

2. Mechanism of

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light response. Checked: phenotype clear, mediator unknown. Matters for whether GO:0007186 could ever be defensible. Resolve by identifying the effector linking RGR photostate to LRAT/hydrolase regulation (G-protein vs conformational/retinoid-transfer).

3. Functional activity of RGR-d at the plasma membrane.

Checked: only localization + complement co-deposit shown. Matters for is_active_in PM. Resolve with activity/interaction assays on RGR-d at the basolateral membrane.

4. Non-ocular (keratinocyte) mechanism

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PMC9014095). Checked (Iteration 2): abstract read in full — RGR framed as a photoisomerase, localized to nucleus/mitochondria/cytoplasm, functional evidence is siRNA-knockdown phenotype only, no G-protein readout. Largely resolved: does not support GPCR signaling. Residual gap: the molecular basis of the knockdown phenotype (retinoid/photoisomerase-dependent vs a novel non-ocular role) remains undefined; would need effector/mechanism studies.

5. Exact GO-term IDs and current review actions in the RGR-ai-review.yaml.

The local gene YAML was not available in this environment; curator should confirm which of the five propagated terms are actually present and their current qualifiers.

Discriminating Tests

1. **Multiplexed G-protein coupling panel** (GTPyS binding, BRET/NanoBiT Gα sensors, TGFα-shedding) on reconstituted human RGR under dark/light — the single most decisive test to accept or reject GO:0004930/0008020/0007186.
2. **DRY/NPxxY restoration mutagenesis** (GRY→DRY, NAINY→NPxxY): if coupling cannot be conferred, reinforces functional divergence.
3. **Isoform-resolved proteomics/interactomics** of full-length RGR vs RGR-d (ER vs plasma membrane) to test whether any PM-localized RGR engages signaling partners vs complement/deposit proteins.
4. **Full-text mining of**

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and any PMC9014095-linked work for a defined non-ocular signaling readout.

5. **Comparative PAINT/annotation review of ancestral node PTN008398221** to flag which descendants genuinely retain coupling vs photoisomerase-only members (RGR, retinochrome, peropsin) — informs whether IBA propagation should be blocked at this clade.

Curation Leads (require curator verification)

- **Action:** Down-annotate / remove the opsin-ancestor-propagated **GO:0004930**, **GO:0008020**, **GO:0007186** for canonical human RGR as **paralog/ancestral over-propagation** (no coupled G protein; DRY+NPxxY absent).
- **Action:** **Generalize/replace GO:0007602 (phototransduction)** with a **visual-cycle / chromophore-regeneration BP**; treat classical phototransduction as non-core (RGR supplies chromophore upstream of, and separate from, the photoreceptor cascade).
- **Action:** **Constrain GO:0005886** to the **RGR-d isoform** (do not assert gene-level `is_active_in` plasma membrane); prefer **ER membrane (GO:0005789)** for the canonical product.

- **Add (leads): MF retinal photoisomerase activity / all-trans-retinal binding; BP visual cycle / retinoid metabolic process (regeneration of 11-cis-retinal)** — better than "protein binding".

- **Candidate references + snippets to verify:**

- [P 39961453](#)
— "While named a G protein-coupled receptor, RGR has no known coupled G protein."
- [P 32493732](#)
— RGR "identified previously as a mammalian retinaldehyde photoisomerase homologous to retinochrome found in invertebrates."
- [P 27011730](#)
— "RGR-d is targeted to the basolateral plasma membrane of the RPE. RGR-d, but not normal RGR ... trafficks to the plasma membrane."
- [P 18474598](#)
— "in the dark, RGR-opsin inhibits lecithin:retinol acyltransferase and all-trans-retinyl ester hydrolase in vitro and that this inhibition is released upon exposure to light" (light-regulated, **no G protein identified**).
- [P 8639565](#)
— RGR binds all-trans-retinal forming pH-dependent photopigments; isolated from bovine RPE microsomes.
- [P 7947717](#)
— "the rgr gene represents the earliest independent branch of the vertebrate opsin gene family"; direct binding shown on **bovine** RGR, human 86% identical by sequence; proposes retinochrome-like isomerization (supports target divergence + photoisomerase MF).
- [P 35445026](#)
— "The retinal G protein-coupled receptor (RGR) is a photoisomerase in the human retina"; non-ocular knockdown phenotype only, intracellular/nuclear localization (does not support GPCR signaling).

- **Suggested questions for curator:** Which of the five terms are actually present with which qualifiers/evidence codes? Is the PM annotation tied to an isoform? Should IBA propagation from PTN008398221 be blocked for the RGR/retinochrome photoisomerase clade?
- **Suggested experiments:** the G-protein coupling panel and isoform interactomics above.

Provenance

Sequence motif analysis executed on UniProt P47804 (291 aa) via `execute_code` (requests + regex): DRY/ERY motif absent (found GRY at position 112); NPxxY absent (found NAINY at position 261). Full abstracts for

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and

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fetches via NCBI E-utilities efetch (Iteration 2) and read in full. Literature retrieved via PubMed search (PMIDs 39961453, 32493732, 37585292, 37057907, 33187985, 18474598, 27011730, 8639565, 7947717, 35445026, 41618564).

Note: the local `RGR-ai-review.yaml` and any `*-bioinformatics` analyses were not present in this environment; exact term IDs/current actions should be confirmed against the repository.
